

University Database – DQL & DML Tasks

DQL

1. Display all student records.

```
SELECT *  
FROM Student;
```

S_id	F_name	L_name	Phone_no	DOB	Department_id	Hostel_id
1	AMANI	ABDULAAH	9779551010	2000-01-01	1	1
2	EBTESAM	OBAID	9779552020	2001-02-02	2	2
3	Ali	AHMED	9779553030	2000-03-03	1	1
4	Sara	Ahmed	9779554040	2002-04-04	3	2

2. Display each student's full name, enrollment date, and current status.

```
ALTER TABLE Student  
ADD EnrollmentDate DATE,  
CurrentStatus VARCHAR(50);
```

	FullName	EnrollmentDate	CurrentStatus
1	AMANI ABDULAAH	2024-09-01	Active
2	EBTESAM OBAID	2024-09-01	Active
3	Ali AHMED	2024-09-01	Active
4	Sara Ahmed	2024-09-01	Active

3. Display each course title and credits.

```
SELECT  
Course_name AS CourseTitle,  
Duration AS Credits  
FROM Course;
```

	CourseTitle	Credits
1	Database Systems	4 months
2	Calculus	6 months
3	Physics I	5 months

4. Display each student's full name and GPA as GPA Score.

```
SQLQuery college.s...BTESAM\ebtes (79)* - X
SELECT
    CONCAT(F_name, ' ', L_name) AS FullName,
    GPA AS [GPA Score]
FROM Student;

ALTER TABLE Student
ADD GPA DECIMAL(3,2);

UPDATE Student
SET GPA = CASE
    WHEN S_id = 1 THEN 3.50
    WHEN S_id = 2 THEN 3.75
    WHEN S_id = 3 THEN 3.20
    WHEN S_id = 4 THEN 3.90
    ELSE NULL
END;

SELECT
    CONCAT(F_name, ' ', L_name) AS FullName,
    GPA AS [GPA Score]
FROM Student;
```

100 %

Results Messages

	FullName	GPA Score
1	AMANI ABDULAAH	3.50
2	EBTESAM OBAID	3.75
3	Al AHMED	3.20
4	Sara Ahmed	3.90

5. List student IDs and names of students with GPA greater than 3.5.

```
SQLQuery college.s...BTESAM\ebtes (79)* - X

SELECT
    S_id,
    CONCAT(F_name, ' ', L_name) AS FullName,
    GPA
FROM Student
WHERE GPA > 3.5;
```

100 %

Results Messages

	S_id	FullName	GPA
1	2	EBTESAM OBAID	3.75
2	4	Sara Ahmed	3.90

6. List students who enrolled before 2022.

```
SQLQuery college.s...BTESAM\ebtes (79)* - X

ALTER TABLE Student
ADD EnrollmentDate DATE;

UPDATE Student
SET EnrollmentDate =
CASE
    WHEN S_id = 1 THEN '2020-09-01'
    WHEN S_id = 2 THEN '2021-01-15'
    WHEN S_id = 3 THEN '2022-05-10'
    WHEN S_id = 4 THEN '2023-08-20'
END;

SELECT
    S_id,
    CONCAT(F_name, ' ', L_name) AS FullName,
    EnrollmentDate
FROM Student
WHERE EnrollmentDate < '2022-01-01';
```

100 %

Results Messages

	S_id	FullName	EnrollmentDate
1	1	AMANI ABDULAAH	2020-09-01
2	2	EBTESAM OBAID	2021-01-15

7. Display students with GPA between 3.0 and 3.5.

SQLQuery college.s...BTESAM\ebtes (79))

```
SELECT
    S_id,
    CONCAT(F_name, ' ', L_name) AS FullName,
    GPA
FROM Student
WHERE GPA BETWEEN 3.0 AND 3.5;
```

100 %

	S_id	FullName	GPA
1	1	AMANI ABDULAAH	3.50
2	3	Ali AHMED	3.20

8. Display students ordered by GPA descending.

SQLQuery college.s...BTESAM\ebtes (79))

```
SELECT
    S_id,
    CONCAT(F_name, ' ', L_name) AS FullName,
    GPA
FROM Student
ORDER BY GPA DESC;
```

100 %

	S_id	FullName	GPA
1	4	Sara Ahmed	3.90
2	2	EBTESAM OBAID	3.75
3	1	AMANI ABDULAAH	3.50
4	3	Ali AHMED	3.20

9. Display the maximum, minimum, and average GPA.

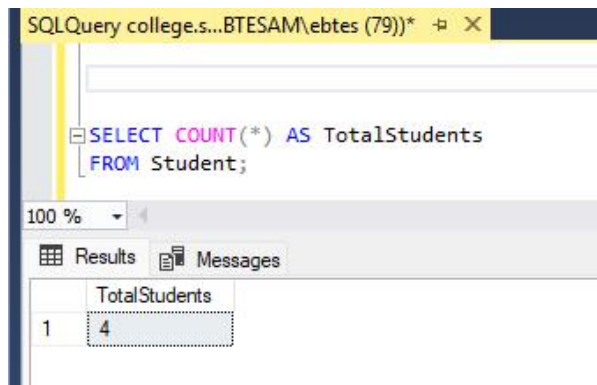
SQLQuery college.s...BTESAM\ebtes (79))

```
SELECT
    MAX(GPA) AS MaxGPA,
    MIN(GPA) AS MinGPA,
    AVG(GPA) AS AvgGPA
FROM Student;
```

100 %

	MaxGPA	MinGPA	AvgGPA
1	3.90	3.20	3.587500

10. Display total number of students.

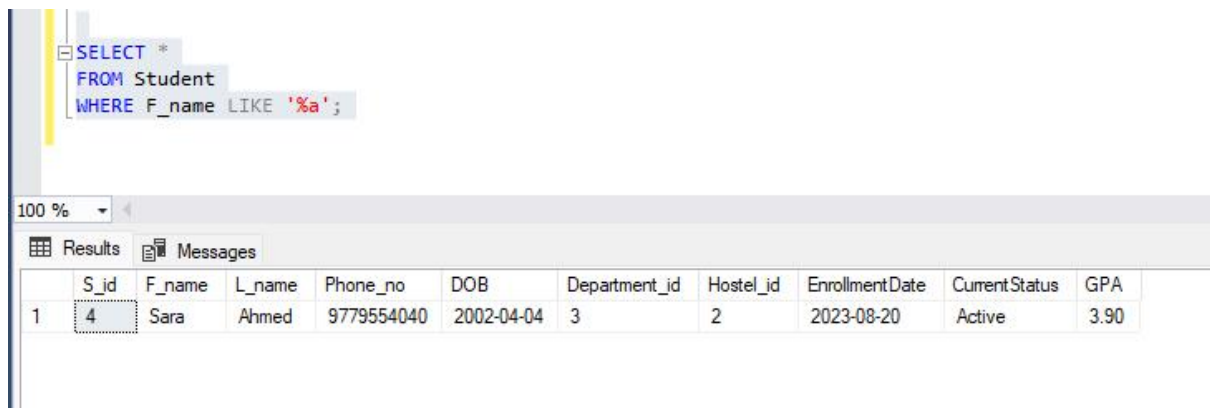


The screenshot shows a SQL query editor window with the title "SQLQuery college.s...BTESAM\ebtes (79)". The query is: `SELECT COUNT(*) AS TotalStudents FROM Student;`. Below the query, the "Results" tab is active, displaying a single row with the column "TotalStudents" and the value "4".

TotalStudents
4

11.

12. . Display students whose names end with 'a'.

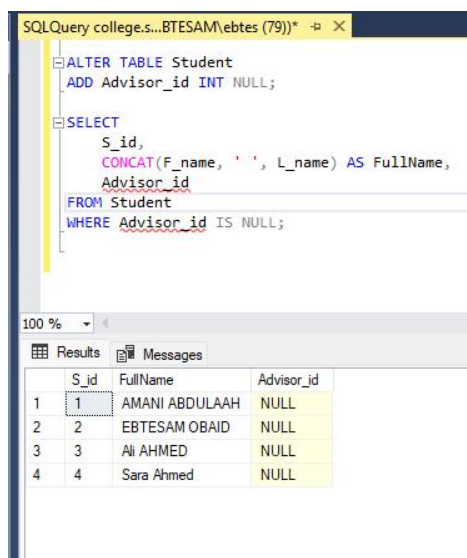


The screenshot shows a SQL query editor window with the title "SQLQuery college.s...BTESAM\ebtes (79)". The query is: `SELECT * FROM Student WHERE F_name LIKE 'a';`. Below the query, the "Results" tab is active, displaying a table with 11 columns: S_id, F_name, L_name, Phone_no, DOB, Department_id, Hostel_id, EnrollmentDate, CurrentStatus, and GPA. The first row shows S_id 4, F_name Sara, L_name Ahmed, and other details.

S_id	F_name	L_name	Phone_no	DOB	Department_id	Hostel_id	EnrollmentDate	CurrentStatus	GPA
4	Sara	Ahmed	9779554040	2002-04-04	3	2	2023-08-20	Active	3.90

13.

12. Display students with NULL advisor.



The screenshot shows a SQL query editor window with the title "SQLQuery college.s...BTESAM\ebtes (79)". The query is: `ALTER TABLE Student ADD Advisor_id INT NULL; SELECT S_id, CONCAT(F_name, ' ', L_name) AS FullName, Advisor_id FROM Student WHERE Advisor_id IS NULL;`. Below the query, the "Results" tab is active, displaying a table with 3 columns: S_id, FullName, and Advisor_id. The first row shows S_id 1, FullName AMANI ABDULAAH, and Advisor_id NULL.

S_id	FullName	Advisor_id
1	AMANI ABDULAAH	NULL
2	EBTESAM OBAID	NULL
3	Ali AHMED	NULL
4	Sara Ahmed	NULL

14.

13. Display students enrolled in 2021.

The screenshot shows a SQL Query Editor window with the following query:

```
SELECT
    S_id,
    CONCAT(F_name, ' ', L_name) AS FullName,
    EnrollmentDate
FROM Student
WHERE YEAR(EnrollmentDate) = 2021;
```

Below the query editor, the 'Results' tab is active, displaying a table with the following data:

	S_id	FullName	EnrollmentDate
1	2	EBTESAM OBAID	2021-01-15

DML

14. Insert your data as a new student (Student ID = 300045, Program ID = 2, GPA = 3.6).

```
SQLQuery college.s...BTESAM\ebtes (79)) * ❏ X
❏ INSERT INTO Student (S_id, ProgramID, GPA)
VALUES (300045, 2, 3.6);
SET IDENTITY_INSERT Student ON;

❏ INSERT INTO Student (S_id, ProgramID, GPA)
VALUES (300045, 2, 3.6);

SET IDENTITY_INSERT Student OFF;

❏ ALTER TABLE Student
ADD ProgramID INT,
GPA DECIMAL(3,2);

SET IDENTITY_INSERT Student ON;

❏ INSERT INTO Student (S_id, ProgramID, GPA)
VALUES (300045, 2, 3.6);

SET IDENTITY_INSERT Student OFF;

❏ SELECT *
FROM Student
WHERE S_id = 300045;
```

100 % ▾ ◀

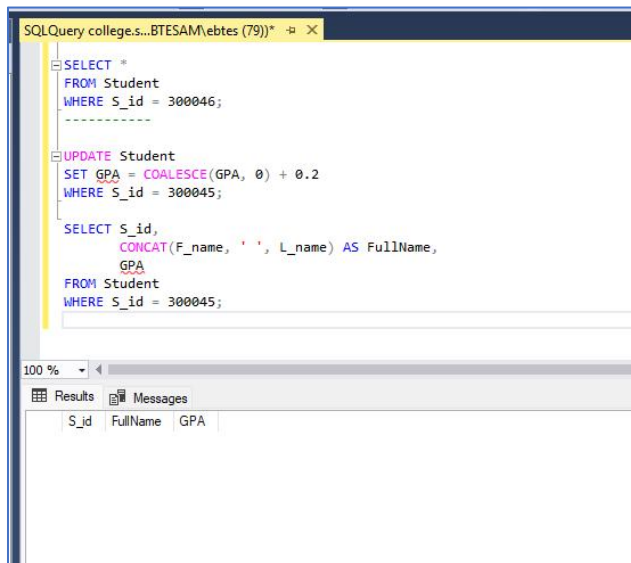
Results Messages

S_id	F_name	L_name	Phone_no	DOB	Department_id	Hostel_id	EnrollmentDate	CurrentStatus	GPA	Advisor_id
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15. Insert another student (your friend) with GPA and advisor set to NULL.

[illegible]

16. Increase your GPA by 0.2.



```
SQLQuery college.s...BTESAM\ebtes (79)) *  X
SELECT *
FROM Student
WHERE S_id = 300046;

UPDATE Student
SET GPA = COALESCE(GPA, 0) + 0.2
WHERE S_id = 300046;

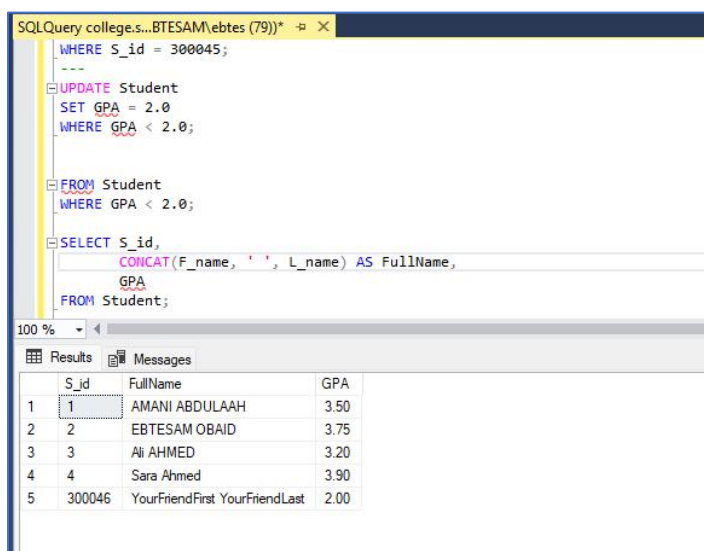
SELECT S_id,
       CONCAT(F_name, ' ', L_name) AS FullName,
       GPA
FROM Student
WHERE S_id = 300046;
```

100 %

Results Messages

S_id	FullName	GPA
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17. Set GPA to 2.0 for students with GPA below 2.0.



```
SQLQuery college.s...BTESAM\ebtes (79)) *  X
WHERE S_id = 300046;
---
UPDATE Student
SET GPA = 2.0
WHERE GPA < 2.0;

FROM Student
WHERE GPA < 2.0;

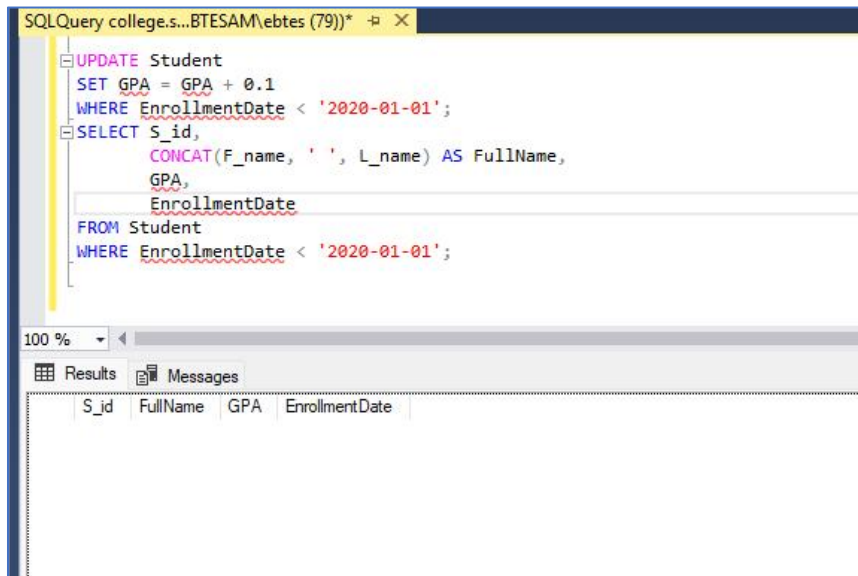
SELECT S_id,
       CONCAT(F_name, ' ', L_name) AS FullName,
       GPA
FROM Student;
```

100 %

Results Messages

	S_id	FullName	GPA
1	1	AMANI ABDULAAH	3.50
2	2	EBTESAM OBAID	3.75
3	3	Ali AHMED	3.20
4	4	Sara Ahmed	3.90
5	300046	YourFriendFirst YourFriendLast	2.00

18. Increase GPA by 0.1 for students enrolled before 2020.



The screenshot shows a SQL Query Editor window with the following SQL code:

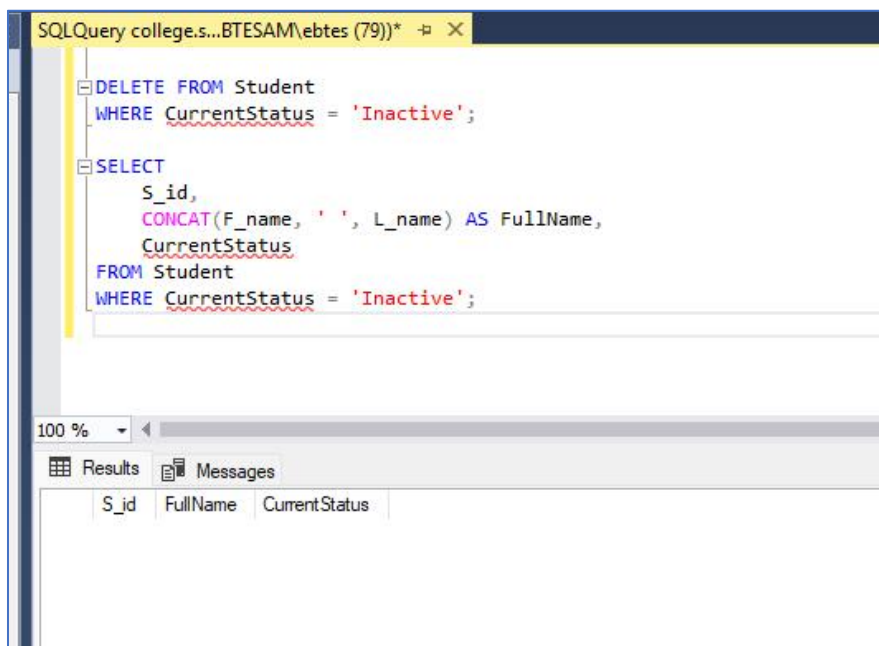
```
UPDATE Student
SET GPA = GPA + 0.1
WHERE EnrollmentDate < '2020-01-01';

SELECT S_id,
       CONCAT(F_name, ' ', L_name) AS FullName,
       GPA,
       EnrollmentDate
FROM Student
WHERE EnrollmentDate < '2020-01-01';
```

Below the code editor, there is a toolbar with 'Results' and 'Messages' buttons. The 'Results' button is active, and a table with the following columns is displayed:

S_id	FullName	GPA	EnrollmentDate
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19. Delete students with status 'Inactive'.



The screenshot shows a SQL Query Editor window with the following SQL code:

```
DELETE FROM Student
WHERE CurrentStatus = 'Inactive';

SELECT
    S_id,
    CONCAT(F_name, ' ', L_name) AS FullName,
    CurrentStatus
FROM Student
WHERE CurrentStatus = 'Inactive';
```

Below the code editor, there is a toolbar with 'Results' and 'Messages' buttons. The 'Results' button is active, and a table with the following columns is displayed:

S_id	FullName	CurrentStatus
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